Year 13 Further Maths - Further vectors

Similar questions - Lines, planes, distances and triple products

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) The plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 14$ and C is the point $(3,1,5)$. Find the shortest distance from C to Π. **[3]**

(b) The lines l_1 and l_2 have equations $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$. Find, in exact form, the shortest distance between l_1 and l_2. **[5]**

2 The points A, B and C have coordinates $(2,1,0)$, $(4,-1,1)$ and $(3,4,-2)$ respectively.

(a) Find a Cartesian equation of the line l through A and B. **[3]**

(b) M is the point on l closest to C. Find the coordinates of M. **[4]**

(c) Find the exact area of triangle ABC. **[4]**

3

(a) Find a vector that is perpendicular to both $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$. **[2]**

(b) Show that the lines $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$ intersect, and state the point of intersection. **[5]**

(c) $\mathbf{a} = 2\mathbf{i} + p\mathbf{j} + (1-p)\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{c} = -2\mathbf{i} + (p-1)\mathbf{k}$. Given that $\mathbf{a} \times \mathbf{b}$ is perpendicular to $\mathbf{c}$, determine the possible values of p. **[6]**

Question 1

(a) $|\mathbf{n}| = \sqrt{4+9+36} = 7$. $\mathbf{n}\cdot\mathbf{c} = 6-3+30 = 33$. Distance $= \dfrac{|33-14|}{7} = \dfrac{19}{7}$

(b) $\mathbf{d}_1 \times \mathbf{d}_2 = (2,6,5)$, magnitude $\sqrt{65}$. $\mathbf{a}_2 - \mathbf{a}_1 = (2,-2,1)$ and the dot product with $(2,6,5)$ is $4-12+5=-3$. Distance $= \dfrac{3}{\sqrt{65}} = \dfrac{3\sqrt{65}}{65}$

Question 2

(a) $\overrightarrow{AB} = (2,-2,1)$, so $\dfrac{x-2}{2} = \dfrac{y-1}{-2} = z$

(b) $M = (2+2t, 1-2t, t)$ and $\overrightarrow{MC} = (1-2t, 3+2t, -2-t)$. Perpendicular to $(2,-2,1)$: $2-4t-6-4t-2-t=0$, so $9t=-6$, $t=-\frac{2}{3}$. $M = \left(\frac{2}{3}, \frac{7}{3}, -\frac{2}{3}\right)$

(c) $\overrightarrow{AC} = (1,3,-2)$. $\overrightarrow{AB}\times\overrightarrow{AC} = (1,5,8)$, magnitude $\sqrt{90} = 3\sqrt{10}$. Area $= \dfrac{3\sqrt{10}}{2}$

Question 3

(a) The cross product is $(-4,7,5)$, i.e. $-4\mathbf{i}+7\mathbf{j}+5\mathbf{k}$ (or any multiple)

(b) Equate: $1+\lambda = 2\mu$, $2-2\lambda = -1+\mu$, $3+\lambda = 7-3\mu$. From the first two, $\lambda = 2\mu - 1$ and $2-4\mu+2 = -1+\mu$ gives $\mu = 1$, $\lambda = 1$. Check the third: $3+1=4$ and $7-3=4$ (true). They intersect at $(2,0,4)$

(c) $\mathbf{a}\times\mathbf{b}\cdot\mathbf{c} = 0$ is the scalar triple product $\begin{vmatrix} 2 & p & 1-p \\ 1 & 2 & 1 \\ -2 & 0 & p-1 \end{vmatrix} = -p^2 - p = 0$, so $p = 0$ or $p = -1$